VISUAL ENCODER BRANCH (Frozen)

Input Image

224 × 224 × 3 RGB

16 × 16 Patching

Patch Partitioning

Patch Extraction & Emb.

N = 196 Patches + [CLS]
Dim: 197 × 768

Patches + Position Enc.

Vision Transformer (ViT-B/16)

12 Transformer Encoder Layers
• Multi-Head Self-Attention (12 heads)
• MLP Blocks (dim = 3072)
• Pre-LayerNorm & Residuals
• 86M Parameters (Frozen)

google/vit-base-patch16-224-in21k

197 × 768 Visual Features

Projection Bridge

Linear (768 → 768) + LayerNorm
Decouples Encoder / Decoder

Visual Memory (Keys & Values) 197 × 768

TRANSFORMER DECODER (Trained from Scratch)

Hidden States h_t (d = 768)

Custom Pre-LN Decoder Layer (× 6 Stack)

d_model = 768, heads = 8, d_ff = 2048, dropout = 0.1

Feed-Forward Network (FFN)

Linear (768 → 2048) → GELU → Linear (2048 → 768)

Multi-Head Cross-Attention (h = 8)

Queries: Text | Keys & Values: Visual Memory
Native extraction of attention matrices α

Masked Multi-Head Self-Attention

Causal Attention Mask (Triangular)

Word Vectors (d = 768)

BERT Tokenizer Embeddings

WordPiece Embedding + Pos. Enc.

bert-base-uncased (Vocab: 30,522, d = 768)

Text Tokens

Caption Tokens (Teacher Forcing)

[CLS], w_1, w_2, ..., w_{t-1} (Max len = 50)

Visual Grounding

Attention Weights α

14 × 14 Spatial Heatmaps
Word-to-Patch Alignment

PREDICTION & DECODING

Tied Linear Layer

Linear Output Projection

Shared weights (768 → 30,522)
Vocabulary Logits

Decoding Strategy

Greedy (11.1 ms/img)
Beam Search k=3 (146 ms/img)
Length Penalty α = 0.7

"A dog running across a green lawn."

Emitted Natural Language Caption